Critical Thresholds in Social Contagion: Phase Transitions Driven by Homophily and Selective Influence

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Critical phenomena, particularly phase transitions where small parameter changes trigger massive cascades, are well documented in physical models. However, mapping criticality in meaning-bearing social systems remains a frontier challenge. While traditional models of collective behavior [4] and social contagion often assume either simplistic random topologies or homogeneous influence, empirical reality is driven by homophily—the tendency of similar individuals to cluster [3]. In this work, we demonstrate that macro-level phase transitions in social adoption are an emergent property inextricably linked to these homophilic networks structures.

We address the scarcity of whole-network data in representative surveys by developing an imputation pipeline. By estimating homophily parameters from ego-centric data (GSS 2004) and imputing plausible network structures to large-scale attitudinal surveys (ATP 2016), we create empirical social topologies. On these networks, we simulate diffusion using a hybrid model: nodes adopt based on either Rational Choice (intrinsic utility) or Selective Social Influence. Our model extends classic threshold models [2] by explicitly treating social proximity as a weight for influence, allowing us to map the system's parameter space.

Using Intrinsic Utility Level (IUL) and Maximum Social Proximity (MSP, the social flexibility to dissimilar influence) as control parameters, our simulations yield pronounced second-order phase transitions. We find a sharp critical threshold where social reinforcement abruptly overcomes local resistance, triggering system-wide cascades (Figure 1). Crucially, this criticality collapses when our "plausible" networks are replaced by degree-preserving Erdős-Rényi baselines, reducing the odds of contagion by 57%. Our results suggest that homophily organizes the network to sustain local reinforcement, acting as the structural prerequisite for criticality in social diffusion.

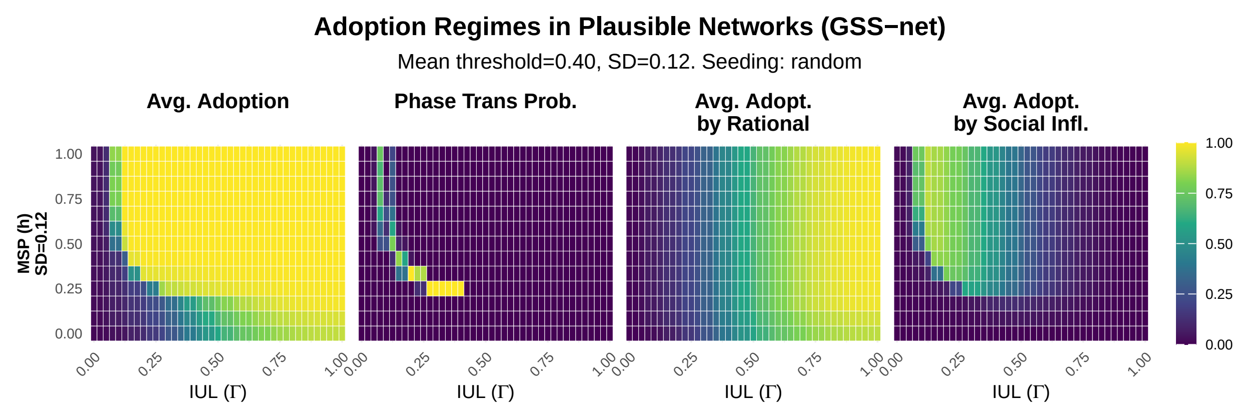

Figure 1: Adoption Regimes. Heatmaps ($SD = 0.12$) showing from left to right: (1) Avg. Total Adoption (Rational + Social), (2) Probability of Phase Transition (abrupt jumps in adoption), (3) Avg. Adoption by Rational Choice, and (4) Avg. Adoption by Social Influence. Axes represent Intrinsic Utility Level (IUL, x-axis) and Maximum Social Proximity (MSP, y-axis). The sharp transition zone (yellow) indicates where social flexibility (h) allows reinforcement to overcome resistance.

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